

China Power Infrastructure

Solid State Transformers – Expert Call Takeaways

We hosted an expert call yesterday on the development of solid state transformers (SST) and high-voltage direct current (HVDC) with the following takeaways: the expert views SST as a potentially important long-term upgrade path for data center (DC) power architecture, given benefits in efficiency, footprint and intelligent/bidirectional power dispatch, but believes commercialization remains early due to immature standards, limited reliability validation, ‘good-enough’ HVDC alternatives, and a lack of proven use cases. He believes near-term adoption will be more likely centered on EV charging, high-energy industrial, renewables+storage, and microgrids, with a broad DC rollout unlikely before 2028. On the value proposition, SSTs can lift end-to-end efficiency by 5+ percentage points vs a line-frequency transformer plus uninterruptible power supply (UPS) chain (typically <90%) and reduce power equipment volume to ~1/5-1/10, while enabling software-defined power management and power-compute coordination. The expert characterized 800V HVDC as an interim architecture that keeps the front-end line-frequency transformer but moves downstream distribution to 800V DC, with UPS, HVDC and SST approaches likely to coexist for 5-10 years, depending on chip power density and DC efficiency targets. Commercial traction is still limited: one electrical equipment company cited ~100-200 North America units, and China XD has four units of overseas orders, but broader DC-scale orders remain scarce. See Table 1 for valuation comps of companies with SST exposure, and Figure 1 for the illustrative diagram.

- SST outlook: benefits and adoption timing:** The expert believes SST offer advantages (energy efficiency, smaller footprint, smarter and bidirectional energy dispatch) and could become a key upgrade path for DC power architectures long term, but commercialization is still early due to immature standards, limited reliability validation, ‘good-enough’ HVDC alternatives, and scarce proven use cases. He believes near-term adoption is more likely in EV charging, high-energy industrial, renewables+storage and microgrids, with large-scale DC rollout unlikely before 2028.
- SST value proposition vs line-frequency transformers:** The expert noted SSTs offer clear advantages over traditional line-frequency transformers in efficiency, footprint and energy dispatch, making them attractive for next-generation AI DC and integrated energy systems. 1) A conventional supply chain built around a line-frequency transformer plus UPS typically delivers less than 90% end-to-end efficiency, while an SST-based approach can lift total system efficiency by more than 5 percentage points, reducing losses, heat and operating costs. 2) In AIDC facilities where transformers and power distribution equipment are taking up more space, SST solutions can cut equipment volume to roughly one-fifth to one-tenth of traditional setups, releasing valuable floor area and supporting higher power density. 3) Unlike traditional transformers that mainly provide one-way power output with limited controllability, SSTs enable software-defined power management and coordination between power supply and computing loads, aligning with smarter DC designs. 4) SSTs can act as an energy router with bidirectional power flow, supporting multi-energy complementarity across wind, solar and storage, while traditional transformers are largely one-directional converters.

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- **800V HVDC architecture: interim approach and evolution path:** The expert believes that the 800V HVDC architecture is best viewed as a transitional pathway. It typically retains the conventional line-frequency transformer on the front end, while shifting the downstream distribution to 800V DC to improve overall system efficiency by reducing conversion losses and current-related inefficiencies. Looking ahead, if SST mature, they could replace the traditional transformer entirely and deliver 800V DC directly, enabling a more efficient end-to-end power conversion chain. Over the next five to ten years, traditional UPS-based architecture, 800V HVDC solutions, and SST-based approaches are expected to coexist. Which design ultimately prevails will depend on the trajectory of chip power consumption and rack power density, as well as each data center's efficiency targets and tolerance for adopting newer power technologies.
- **Commercial traction: early signs and current limitations:** The expert believes that SST commercialization remains at an early stage. An electrical equipment company has indicated it has secured roughly 100-200 units of orders in North America, which is a notable proof point, while China XD has won four units of overseas orders. Apart from these, there is still limited evidence of clear, sizable commercial orders specifically for data center deployments. More than 20 companies in China have already announced SST products, and over 30 more are in pre-release, yet most remain in development and lack true mass-production capability. Overseas players such as Eaton (ETN US, NR) have also announced products, but the expert hasn't seen widely referenced commercialization cases. Overall, meaningful large-scale SST adoption is more likely after 2028, when the market may enter a small-batch, scaled commercial phase; until then, deployments are expected to focus on pilots, niche scenarios, and operational learning.
- **Commercialization constraints: standards, validation, and economics:** The expert believes that SST commercialization is constrained by four main factors. First, there is a lack of finalized specifications and standards, and the broader data center supply chain still lacks unified product standards, which delays both vendor roadmaps and owner investment decisions. Second, reliability validation remains insufficient: data centers require extremely high power availability, but operating data and long-duration field proof for SSTs are still limited, making owners reluctant to approve large-scale deployments. Third, today's 800V HVDC solutions are 'good enough' for current needs, meaning that SSTs can be seen as over-engineered in some use cases and therefore harder to justify economically or operationally. Fourth, rising silicon carbide device prices are putting upward pressure on system costs; however, cost is not viewed as the primary blocker. The more fundamental issue is the limited willingness among owners to pilot SSTs at scale, given uncertain standards, incomplete reliability evidence, and an unclear set of near-term applications where SSTs deliver uniquely compelling benefits.
- **Vendor landscape: positioning and key differentiators:** Among SST vendors, the expert believes that Delta Electronics (2308 TT, covered by Albert Hung, OW) is generally viewed as a first-tier player, having started R&D earliest and achieved the strongest industry recognition, with its key edge coming from early deployments and accumulated operating data. Some other Chinese players may also stand out, including Sungrow (see [note](#) for Sungrow's launch of SST products by our China renewable analyst). Sifang Automation (601126 CH, NC) has a utility-grid background and practical experience in PV-storage-DC architectures, with relatively advanced industrialization, and is exploring cooperation with a leading US DC electrical equipment supplier, aiming to leverage the partner's North American channel presence while it works toward the necessary certifications, according to the expert. Overall, domestic vendors are not clearly disadvantaged on the hardware front; the larger gap is access to real deployments and operating data, where Delta leads.

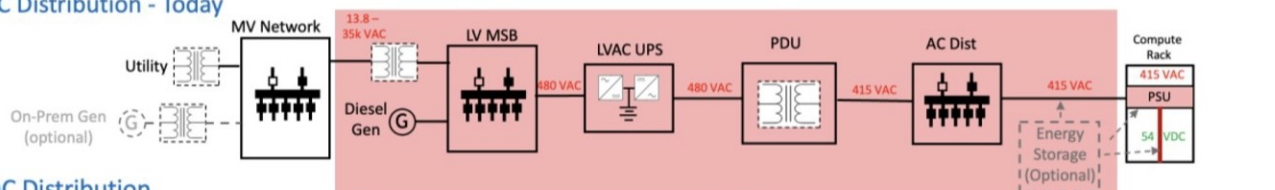
Table 1: Companies with SST exposure

	Ticker	JPM rating	Share price (LC)	Mkt Cap (USDm)	Daily liquidity (USDm)	PE (x)		P/BV (x)		ROE (%)		EV/EBITDA (x)	
						2026E	2027E	2026E	2027E	2026E	2027E	2026E	2027E
Mainland China													
Qingdao TGOOD	300001 CH	OW	30.5	4,747	197.9	20.8	17.0	3.3	2.8	16.8	17.8	14.2	12.3
Sungrow	300274 CH	OW	101.5	31,026	1,711.7	13.8	11.4	3.5	2.8	26.7	25.6	10.9	9.1
China XD	601179 CH	NC	12.1	9,129	472.1	39.7	32.3	2.5	2.4	6.5	7.6	21.0	18.1
Sifang Automation	601126 CH	NC	49.8	6,113	323.6	42.2	36.0	7.7	7.0	18.7	20.0	32.7	27.7
Sinexcel Electric	300693 CH	NC	36.9	1,704	143.8	19.3	15.5	4.3	3.5	24.4	24.8	15.5	12.4
Eaglerise Electric	002922 CH	NC	25.9	1,615	148.5	22.9	15.1	2.6	2.4	11.9	16.4	14.5	10.8
Korea													
LS Electric	010120 KS	N	176,800	17,856	474.3	52.2	37.6	10.6	8.7	22.2	25.4	31.1	23.0
Hyosung Heavy	298040 KS	OW	2,522,000	15,815	220.6	29.6	20.7	7.6	5.7	29.2	31.6	21.1	15.0
Others													
GE Vernova	GEV US	OW	1,042.6	280,167	2,893.7	31.3	38.2	16.4	12.0	39.8	36.1	43.2	26.8
ABB	ABBN SW	N	83.8	187,851	241.0	33.2	32.2	11.0	9.2	33.8	31.0	21.7	21.0
Schneider Electric	SU FP	OW	268.6	176,607	297.1	25.1	21.6	5.6	5.0	24.0	25.0	17.0	14.8
Eaton	ETN US	NR	402.9	156,427	1,045.5	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Delta Electronics	2308 TT	OW	1,800.0	145,209	833.8	40.3	26.5	12.8	9.8	31.8	36.9	25.2	17.1
Vertiv	VRT US	NR	305.9	117,487	1,979.3	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Enphase Energy	ENPH US	N	43.1	5,675	381.1	22.5	20.4	4.3	3.5	20.9	18.9	16.7	15.5
SolarEdge	SEDG US	N	52.1	3,170	197.1	179.8	23.4	5.7	3.9	3.6	20.1	45.5	15.8

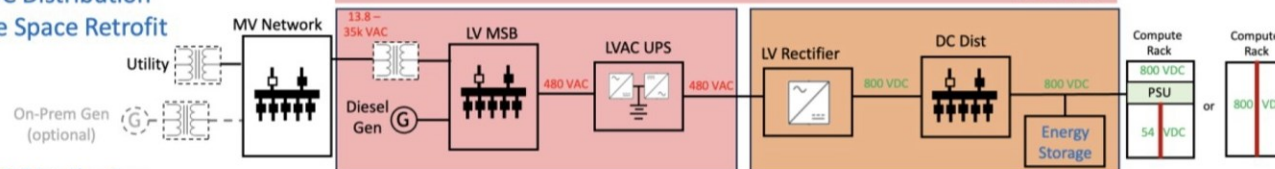
Source: Bloomberg Finance L.P. (consensus estimates for not covered [NC] names), J.P. Morgan estimates. Priced as of 14 Jul 2026.

Figure 1: Datacenter architecture over time

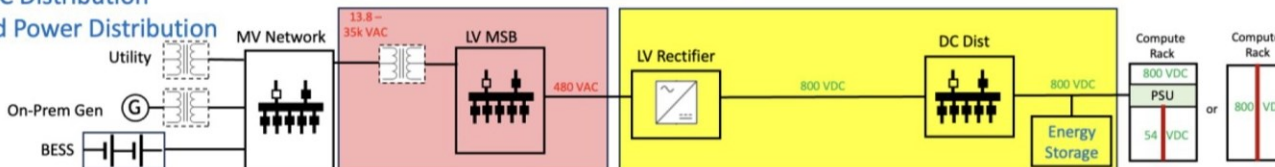
415 VAC Distribution - Today



800 VDC Distribution – White Space Retrofit



800 VDC Distribution - Hybrid Power Distribution



800 VDC Distribution - MV to 800 VDC



Source: NVIDIA.

Companies Discussed in This Report (all prices in this report as of market close on 14 July 2026, unless otherwise indicated)
Delta Electronics, Inc.(2308.TW/NT\$1,855.00/OW), Sungrow - A(300274.SZ/Rmb108.00[13 July 2026]/OW)

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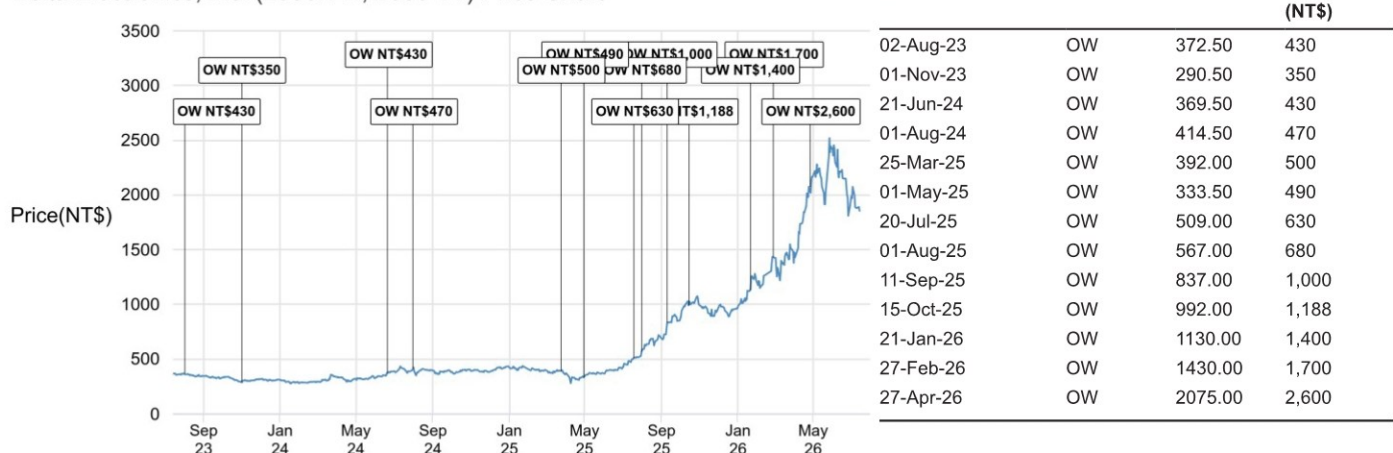
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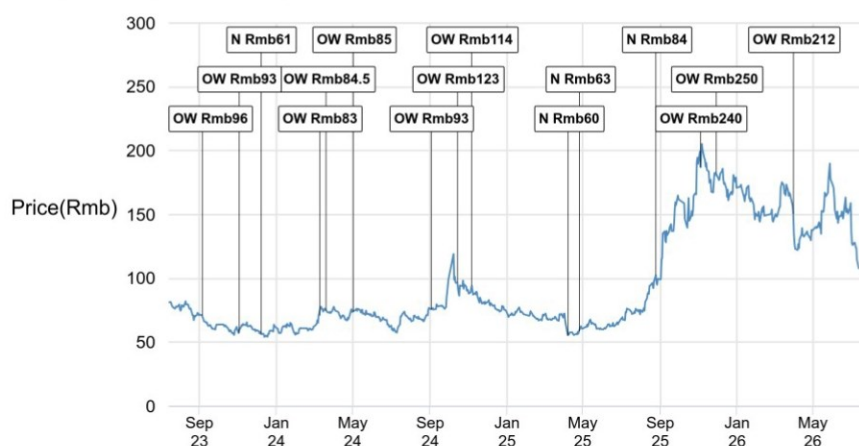
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Delta Electronics, Inc. (2308.TW, 2308 TT) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
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Sungrow - A (300274.SZ, 300274 CH) Price Chart



Date	Rating	Price (Rmb)	Price Target (Rmb)
06-Sep-23	OW	70.93	96
03-Nov-23	OW	57.36	93
08-Dec-23	N	56.79	61
10-Mar-24	OW	71.00	83
21-Mar-24	OW	74.98	84.5
03-May-24	OW	73.82	85
03-Sep-24	OW	75.98	93
15-Oct-24	OW	96.75	123
07-Nov-24	OW	94.12	114
09-Apr-25	N	55.71	60
27-Apr-25	N	58.82	63
26-Aug-25	N	102.60	84
05-Nov-25	OW	187.19	240
30-Nov-25	OW	182.90	250
01-Apr-26	OW	150.76	212

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
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